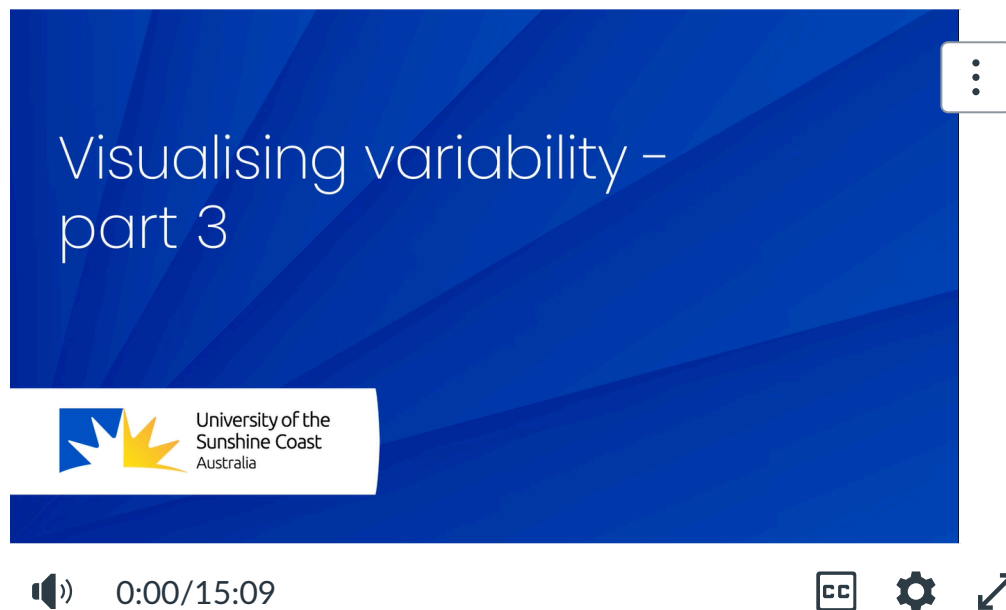


6.3 Statistical analysis of distributions of quantitative variables ^A↓

□ Watch: Visualising variability (part 3) [15.09 mins]

The video below defines and explains terms that are used to measure central location and variability. Box and whisker charts are also described. This is useful as it extends understanding of variability and the ways in which it can be visualised.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138646?wrap=1>

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Visualising variability \(part 3\)](https://learn.usc.edu.au/courses/35242/files/3138666?wrap=1)
<https://learn.usc.edu.au/courses/35242/files/3138666?wrap=1>

Measures of location

The term “measure of location” usually refers to a single “central” value that is considered to characterise a set of values. In other words, it is a “central” value around which all the other values of the set are distributed. There are three measures of location that are more commonly used – mean, median and mode:

- The *mean* is the average value for a set of values. It is influenced by outlying values (i.e. extremely small and extremely large values).
- The *median* is the value in the middle of the set when the values are listed in ascending order. This means that when there are an odd number of values, it will be the number that is in the middle. However, when there are an even number of values, the two values in the middle must be added together and the average of the sum of those two numbers is calculated to be the median value. The median is often the preferred measure of location, as it is not influenced by outliers. When data is severely skewed, it is better to use the median as the measure of central location.
- The *mode* is the value that occurs most frequently in the set of values. More than one mode can exist in a set of values, as the same number of instances of different values can exist in the set. Where no values occur more than once in the dataset, the set of values has no mode. The mode can be useful as a measure of location when the data has a small set of distinct values. When the dataset is large, a histogram can be used to identify the mode – the tallest column will represent the mode.

Measures of variability

In variations, there are also “measures of variability”, key terms of which will be defined below.

Range is the simplest measure of variation. The highest number in a set of values minus the lowest number in the set will give you the range. The range provides a measure of variability. However, because range only relies on two values (the highest and the lowest), it may be influenced by extreme values. Therefore, normally further measures of variability are required. These include:

- *Standard deviation* based on how dispersed the data values are in relation to the mean – how much each value deviates from the mean. It is reliable when a histogram of the values has a symmetric or bell-shaped distribution. However, it is influenced by extreme values and therefore not useful for skewed data.
- *Percentiles* indicate the value below which a given percentage of observations within a group of observations falls. Common percentiles are the 25th, 50th and 75th percentiles (also called quartiles).
- *Quartiles* are the 25th, 50th and 75th percentiles. They are called quartiles because they divide the data into four parts/quarters.
- *Interquartile range* is the middle 50% of the distribution of a variable's values – i.e. the difference between the third quartile (75th percentile) and first quartile (25th percentile). It is advantageous to use percentiles and the interquartile range, because:
 - extreme values do not distort the value of percentiles
 - a bell-shaped distribution is not required for a percentile to accurately convey variability.

□ Activity: Check your understanding – Measures of location and variability

In the activity below, can you drag the definitions into the correct row?

You have made 2 attempts. You got 87.5% correct on your last attempt.



Range

The simplest measure of variation

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Standard deviation

Based on how dispersed the data values are in relation to the mean



Influen
therefo

Percentile

Value b
percen

Quartile

Data divided into four parts



Interquartile range

Middle 50% of the distribution of a variable's values



Mean

average value for a set of values



Median

Value in the middle of a set of values when they are listed in ascending order

Better
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Mode

Value that occurs most frequently in a set of values



Useful
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7/8

↺ Retry

Box and whisker charts

Box and whisker charts show a visual summary of the distribution of the values of a variable by illustrating the quartiles. They allow a detailed comparison between the distributions of multiple variables.

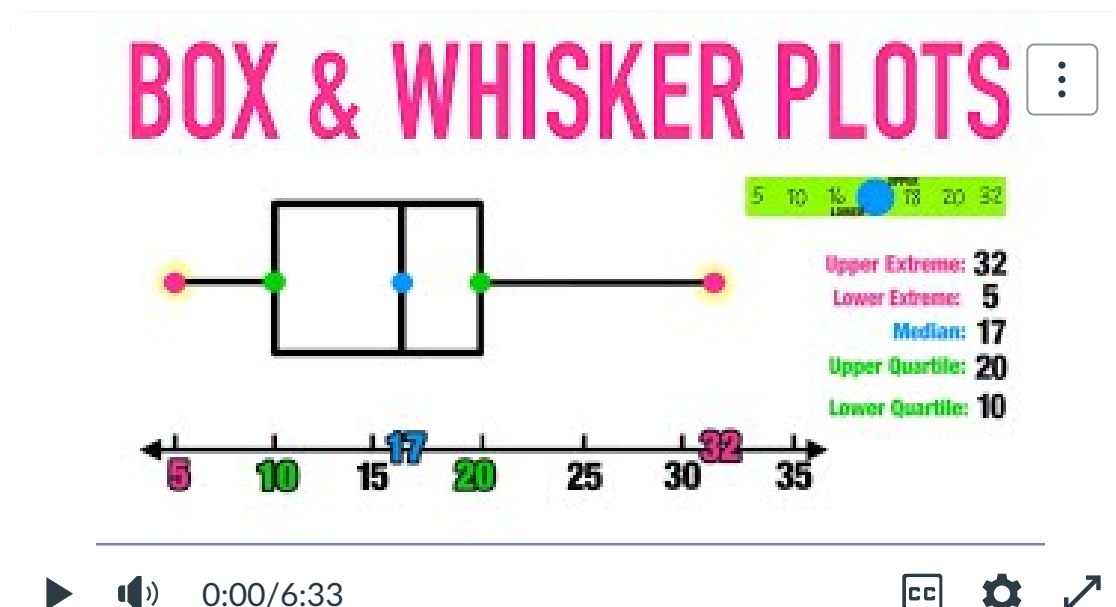
- The ends of each box represent the first and third quartiles.
- The line dividing the box represents the median.
- The lines (whiskers) extend from each end of the box.
 - One line extends to the highest value that is less than or equal to the third quartile.
 - The other line extends to the lowest value that is greater than or equal to the first quartile.
- The mean value is marked with an X.
- A value outside the range is considered an outlier and is marked by a dot.

□ Watch: Box and whisker plots explained [6.32 mins]

This video is useful as it describes what box and whisker plots are and how they can be used. It also explains terms such as *upper quartile*, *lower quartile* and *interquartile range*.

As you watch, consider:

- How can box and whisker charts improve the understanding of the distribution of a dataset?
- Why and when could this be useful for visualising data?



Source: © Mashup Math (2015).

□ Activity: Democracy report 2022

Read and reflect on the report by the V-Dem Institute below. As you read, take time to consider and make notes on:

- How easy is it to comprehend the whole report in one reading?
- How could you visualise the content of this report in a short presentation?
- What techniques would you use, and why?

Ensure that you can justify your thinking and bring your answer to the workshop.

V-Dem Institute. (2022). [Democracy report 2022: Autocratization changing nature?](https://v-dem.net/media/publications/dr_2022.pdf)  (https://v-dem.net/media/publications/dr_2022.pdf)